B.Tech. BCSE497J - Project-I

PRIVACY-PRESERVING MEDICAL DIAGNOSIS USING FEDERATED LEARNING

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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(Deemed to be University under section 3 of UGC Act, 1956)

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DECLARATION

I hereby declare that the project entitled **Privacy-Preserving Medical Diagnosis using Federated Learning** submitted by me, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* to VIT is a record of bonafide work carried out by me under the supervision of Dr. Adaline Suji.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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This is to certify that the project entitled **Privacy-Preserving Medical Diagnosis using Federated Learning** submitted by **Uddhav Sethi (23BKT0092)**, **School of Computer Science and Engineering**, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by the candidate under my supervision during Fall Semester 2026-2027, as per the VIT code of academic and research ethics.

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List of Abbreviations

AUPRC	Area Under the Precision-Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
DICOM	Digital Imaging and Communications in Medicine
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
ECE	Expected Calibration Error
FedAvg	Federated Averaging
FL	Federated Learning
gRPC	Google Remote Procedure Call
Grad-CAM	Gradient-weighted Class Activation Mapping
IoU	Intersection over Union
MC Dropout	Monte Carlo Dropout
MLflow	Machine Learning Lifecycle Platform (experiment tracker)
NPV	Negative Predictive Value
OOD	Out-of-Distribution
RDP	Renyi Differential Privacy
RSNA	Radiological Society of North America
SecAgg+	Secure Aggregation (Bell et al. variant)
TLS	Transport Layer Security
VRAM	Video Random Access Memory

Symbols and Notations

ε	Privacy budget (epsilon) of differential privacy
δ	Failure probability (delta) of differential privacy; fixed at 10^{-5}
C	Per-sample gradient clipping norm (max_grad_norm = 1.0)
σ	Noise multiplier of the Gaussian mechanism
B	Batch size
n_k	Number of training samples held by hospital k
w_k	Head parameters returned by hospital k
T	Number of stochastic forward passes in MC Dropout (T = 20); also the softmax temperature in Chapter 7
$\bar{p}$	Mean predictive distribution over the T passes
$H(\bar{p})$	Predictive entropy of $\bar{p}$
τ	Decision threshold on P(pneumonia)
α	Dirichlet concentration (non-IID severity); also the conformal miscoverage level

ABSTRACT

Deep learning models for chest X-ray analysis can reach radiologist-level accuracy, but they are normally trained on images pooled from many hospitals, something that privacy law and institutional policy often prohibit. Federated learning keeps the images where they are, yet the model updates that hospitals exchange can still leak information about individual patients, and a prediction with no explanation or confidence estimate is of limited use to a clinician. This project builds and measures an end-to-end pneumonia detection pipeline that addresses all three concerns together.

Three simulated hospitals (the Kermany chest X-ray set as one site and two shards of the RSNA Pneumonia Detection data as the others) train a shared DenseNet121 classifier with Flower. The ImageNet-pretrained backbone is frozen with its BatchNorm statistics fixed, and only a head of about 263 thousand parameters is trained and exchanged. This choice makes per-sample differential privacy (Opacus DP-SGD with an RDP accountant, reported as epsilon and delta) applicable, shrinks each update to about 1 MB, and keeps secure aggregation affordable. Updates are masked with Flower's SecAgg+ so the server only sees their sum, and all traffic runs over gRPC with TLS and per-hospital client authentication. Predictions carry Grad-CAM heatmaps and Monte Carlo Dropout uncertainty, and the least certain cases are deferred to a human reviewer.

Because each technique is already known, the contribution of the work is the integration and the measured cost of each layer, reported as an ablation over three seeds. On the pooled test set, centralized training reaches an AUROC of 0.905, federated averaging 0.814, adding secure aggregation 0.819, and adding differential privacy at epsilon = 4 (delta = 10^{-5}) 0.809, with a clean monotonic privacy-utility curve from epsilon = 1 to 8. Differential privacy mainly costs sensitivity and roughly quadruples calibration error. The deployed model, run at a validation-selected decision threshold of 0.35, reaches an AUROC of 0.80 (mean over three seeds), with sensitivity and specificity near 73 percent; on the subset of images with unambiguous labels the same model reaches an AUROC of 0.88. A loss-based membership-inference attack finds no exploitable signal at any epsilon. The pipeline runs both as a simulation and as a Docker deployment with real TLS, and is backed by 255 automated tests.

Keywords: federated learning, differential privacy, secure aggregation, pneumonia detection, chest X-ray, Grad-CAM, Monte Carlo Dropout, DenseNet121

CHAPTER 1

1. INTRODUCTION

1.1 Background

Pneumonia is one of the leading infectious causes of death worldwide, and chest radiography is the first imaging test most patients receive for it. Reading a chest X-ray correctly takes training, and in many places the number of radiologists is far below the number of scans. Convolutional networks trained on large labelled collections have shown that a model can flag pneumonia on a frontal chest film at a level comparable to practising radiologists [1], which makes such models attractive as a second reader or a triage aid.

The difficulty is where the training data comes from. A model improves with more and more varied examples, and the examples sit in separate hospitals, each bound by consent rules, national health-data law and its own institutional review. Copying the images to one central server is often impossible. Federated learning offers a way around this: each hospital trains the model on its own images and sends only the resulting model parameters to a coordinator, which averages them into a shared model [2]. The approach has been argued to be a practical route to multi-institution medical AI [3, 4, 5].

Federation alone does not make a system private. Parameters computed from patient images carry information about those images, and published attacks show that membership of a record in the training set, and in some settings the record itself, can be recovered from gradients or model outputs [6, 7]. Meaningful protection therefore needs further layers: a formal privacy mechanism applied to the update, a cryptographic guarantee that the coordinator cannot read any single hospital's update, and a secure channel that also authenticates who is sending it. Finally, a clinician is unlikely to act on a bare label. The system should show which part of the film drove the prediction and should say when it is unsure.

1.2 Motivations

Most published pneumonia detectors are trained on a dataset that has already been pooled, so the data-sharing problem that motivates federated learning is assumed away. Work that does use federation frequently stops at averaging the weights and treats the result as private. Studies that add differential privacy or secure aggregation usually do so on a small convolutional network and rarely report what the protection costs in accuracy, calibration, explanation quality or communication. A hospital deciding whether to adopt such a system needs exactly those numbers.

Three practical observations drove the design of this project. First, a standard DenseNet121 is built on batch normalisation, which mixes information across the samples of a batch and therefore breaks the per-sample gradient computation on which differential privacy depends. Second, a full DenseNet121 update is about 28 MB, which is heavy to mask and encrypt every round and does not fit the default gRPC message limit. Third, the project had to run on a single laptop GPU with 4 GB of memory. All three point toward

the same remedy, training and exchanging only a small head on top of a frozen pretrained backbone, and the report measures what that remedy gives up.

The project also supports the United Nations Sustainable Development Goal 3, good health and well-being, by showing a route to collaborative diagnostic models that do not require patients to give up control of their images.

1.3 Scope of the Project

The scope covers a prototype for binary classification (pneumonia versus normal) of frontal chest X-rays, evaluated on public data with simulated hospitals. In scope are: the federated training pipeline; sample-level differential privacy with a formal accountant; secure aggregation; transport security with client authentication; Grad-CAM explanations; Monte Carlo Dropout uncertainty with a deferral path to human review; an out-of-distribution check and an input gate that rejects images that are not chest X-rays; and a reproducible evaluation that measures each layer's cost through an ablation ladder.

Out of scope for this phase are malicious or poisoning participants, collusion above the secure-aggregation threshold, side-channel and physical attacks, a real multi-hospital deployment, integration with hospital image archives, and any claim of clinical validity. The hospitals are processes and containers on one machine. The tool is a research prototype and is not a medical device.

The remainder of the report is organised as follows. Chapter 2 reviews the literature, states the gap and the objectives, and gives the project plan. Chapter 3 lists requirements, feasibility and the hardware and software used. Chapter 4 presents the architecture and the design diagrams. Chapter 5 describes each module and how it was tested. Chapter 6 shows the system running, Chapter 7 reports and discusses the results including cost, and Chapter 8 concludes.

CHAPTER 2

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

Deep learning on chest X-rays. DenseNet connects every layer to all later layers in a block, which encourages feature reuse and gives strong accuracy with relatively few parameters [8]. CheXNet, a 121-layer DenseNet trained on the ChestX-ray14 collection, reported pneumonia detection on a par with radiologists [1], and DenseNet121 has since become the usual baseline for chest radiograph work. Kermany et al. released a labelled set of paediatric chest X-rays from Guangzhou with bacterial and viral pneumonia and normal classes [9]. The RSNA Pneumonia Detection Challenge supplied an adult set with expert bounding boxes around lung opacities, derived from the NIH collection [10]. The two sources differ in age group, scanner and labelling convention, which is useful here because it gives the simulated hospitals genuinely different data.

Federated learning. Federated averaging (FedAvg) alternates local training with a weighted average of the returned parameters [2]. Kairouz et al. survey the open problems, among them statistical heterogeneity, communication cost and privacy [11]. Heterogeneity is the issue that matters most for hospitals: when local label distributions differ, the average can fit none of them well, and batch-normalisation statistics are especially affected [12]. Group normalisation is the usual replacement because it does not depend on batch statistics [13]. Flower is a framework that runs the same client and server code in simulation or as separate processes [14].

Differential privacy. A randomised mechanism M is (ε, δ)-differentially private if its output distribution changes by at most a factor e^{ε} (plus δ) when a single training record is added or removed [15]. DP-SGD makes neural network training satisfy this definition by clipping each per-sample gradient to a norm C and adding Gaussian noise to the sum, with a moments accountant tracking the cumulative privacy loss [16]. Renyi differential privacy gives tighter composition across many steps [17], and Opacus implements DP-SGD for PyTorch with an RDP accountant [18]. A requirement that follows directly from the definition is that the loss on one sample must not depend on other samples in the batch, which is why Opacus rejects models containing batch normalisation.

Secure aggregation. In the protocol of Bonawitz et al., every pair of clients agrees on a random mask that one adds and the other subtracts; the masks cancel in the sum, so the server learns the aggregate but no individual update, and secret sharing lets the round complete when some clients drop out [19]. Bell et al. reduce the communication and computation cost by having each client talk to a small random subset of the others, which is the variant Flower ships as SecAgg+ [20]. The protocol works on integers in a finite field, so real-valued updates must be quantised first.

Explanation, uncertainty and calibration. Grad-CAM weights the feature maps of a chosen convolutional layer by the gradient of the class score and produces a coarse heatmap of the regions that influenced the prediction [21]. Gal and Ghahramani showed

that keeping dropout active at test time and averaging several stochastic passes approximates Bayesian inference and gives an uncertainty estimate [22]. Modern networks tend to be overconfident, and temperature scaling on a validation set is a simple, effective correction [23]. Conformal prediction gives distribution-free coverage guarantees on prediction sets [24, 25], and isolation forests detect anomalous inputs from tabular feature vectors [26]. Contrast limited adaptive histogram equalisation (CLAHE) improves local contrast in radiographs [27].

The techniques used in this project and the part each plays are summarised in Table 2.1.

Table 2.1 Techniques used in the project, their sources and their role

Technique	Source	Role in the pipeline
DenseNet121 (ImageNet weights)	Huang et al. [8]	Frozen feature extractor
FedAvg / Flower	[2, 14]	Orchestration and aggregation
DP-SGD, RDP accountant, Opacus	[16, 17, 18]	Formal (ε, δ) protection of the update
SecAgg+	[19, 20]	Server sees only the sum of updates
TLS with client authentication	Standard protocol	Confidentiality, integrity, identity
Grad-CAM	[21]	Visual explanation
MC Dropout, temperature scaling	[22, 23]	Confidence and deferral
Isolation forest, conformal sets	[25, 26]	Input checks and coverage analysis
CLAHE	[27]	Contrast preprocessing

2.2 Research Gap

Each of the components above has been studied on its own, and individual combinations appear in the literature, for example federated learning with differential privacy or with secure aggregation. We did not find a single open, reproducible pipeline for pneumonia detection that combines federated learning, sample-level differential privacy with a stated (ε, δ), secure aggregation, authenticated transport, explanation and uncertainty, and that also reports what each layer costs. Where such combinations are described, the protection is often asserted without an accountant, or the cost is reported for accuracy alone. This project does not propose a new algorithm. Its contribution is the integration, an ablation that isolates the price of each layer, and an honest account of the interactions between layers.

2.3 Objectives

The project was set seven objectives, in the order in which they take priority when they conflict (privacy before security before correctness before reproducibility, with raw accuracy last):

- Build a federated pipeline in which hospitals train locally and share only model updates, never patient images.

- Apply differential privacy with a formal mechanism and accountant, and report privacy as (ε, δ).
- Add secure aggregation so that the server observes only the aggregate of the hospitals' updates.
- Secure all communication with TLS and client authentication, so that eavesdropping, tampering and impersonation are prevented.
- Reach strong pneumonia detection performance with DenseNet121 transfer learning despite limited per-hospital data.
- Make the model explainable with Grad-CAM and confidence-aware with Monte Carlo Dropout, deferring low-confidence cases to a human.
- Produce a reproducible artifact: seeded, configuration-driven, tracked and tested.

2.4 Problem Statement

Given chest X-ray collections held by several institutions that cannot exchange images, train a pneumonia classifier collaboratively such that (i) no image leaves its institution, (ii) no single patient's data can be inferred from what is shared, to a stated (ε, δ), (iii) the coordinating server cannot read any institution's individual update, (iv) the channel between the parties is encrypted and mutually authenticated, and (v) each prediction is accompanied by an explanation and an uncertainty estimate that lets uncertain cases be referred to a clinician. Quantify, for every layer added, the change in accuracy, calibration, explanation quality, communication and compute.

2.5 Project Plan

Work was organised as a sequence of small, individually testable stages, each ending with a passing test or a live run before the next began. Table 2.2 lists the phases. Decisions that changed the approved scope, such as the dataset choice, the label harmonisation and the privacy budget, were taken as explicit decision gates and recorded.

Table 2.2 Project phases and stages

Phase	Stages	Outcome
0 Foundation	0–2	Repository, pinned Python 3.11 environment, Hydra configuration, seeding, local MLflow tracking
1 Data	3–7	Dataset download with checksum verification, patient-level splits, three-hospital partition, CLAHE cache, transforms
2 Model and baselines	8–12	Frozen DenseNet121 with trainable head, feature cache, metrics, local and centralized baselines
3 Federated and privacy	13–17	FedAvg, DP-SGD with accountant, SecAgg+, TLS and client authentication, Docker Compose deployment
4 Clinical trust	18–19	Grad-CAM explanations, Monte Carlo Dropout and deferral
5 Evaluation and delivery	20–23	Overhead instrumentation, 27-run ablation campaign, test-suite hardening, documentation
6 Extensions	OPT-1 to OPT-6	Calibration, privacy attack, Grad-CAM localisation, conformal sets, OOD gate, demo application

CHAPTER 3

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

- **FR1.** Load chest X-rays in JPEG, PNG or DICOM format, apply CLAHE and the standard transforms, and keep every image on the filesystem of the hospital that owns it.
- **FR2.** Train the classifier head locally at each hospital for a configurable number of epochs, with optional DP-SGD.
- **FR3.** Quantise and mask each update with SecAgg+ before it leaves the hospital, and have the server aggregate with FedAvg.
- **FR4.** Authenticate every hospital to the server and encrypt all traffic.
- **FR5.** Report privacy as (ε, δ) from an accountant, and report accuracy as AUROC with sensitivity, specificity, F1 and calibration, as mean and standard deviation over three seeds.
- **FR6.** Produce a Grad-CAM heatmap and a Monte Carlo Dropout confidence for a given image, and flag the least certain cases for human review.
- **FR7.** Reject inputs that are not chest X-rays, and flag images unlike a hospital's training data.
- **FR8.** Run the identical client and server code in simulation and in a multi-container deployment.
- **FR9.** Make every privacy layer independently switchable so that the ablation ladder can be produced.

3.1.2 Non-Functional

- **Privacy and security.** No image and no unmasked individual update may cross a hospital boundary. Certificates and keys are generated by script and are never committed.
- **Reproducibility.** Every run is defined by a configuration, seeded for torch, numpy, random, data partition and client sampling, and logged to MLflow with the resolved configuration. Dependencies are pinned exactly.
- **Resource limits.** The full system must run on a laptop with a 4 GB GPU, with clients executed sequentially.
- **Communication.** The per-round payload must stay far below gRPC's message ceiling.
- **Testability.** Every privacy-relevant claim is backed by an automated test or an accountant output.

- **Honesty of reporting.** Limitations and negative findings are reported alongside positive results.

3.2 Feasibility Study

3.2.1 Technical Feasibility

Every component is available as a maintained open-source Python library, and the combination was demonstrated before the ablation campaign started. The one real obstacle was memory: Opacus stores per-sample gradients, and these cannot be computed for the full DenseNet121 in 4 GB. Freezing the backbone removes the obstacle because the head has only about 263 thousand parameters. A later experiment that unfreezes the last dense block (about 2.4 million trainable parameters) peaked at 441 MB of GPU memory under DP-SGD, so the approach remains feasible with a larger trainable part. The Flower version was pinned exactly because the framework's API changed across 1.x releases and Secure Aggregation is available only through its legacy workflow interface at the pinned version.

3.2.2 Economic Feasibility

The project has no licence cost: all software is open source and both datasets are free for research use. Development used a personal laptop with a consumer GPU, and the demonstration application is hosted on a free community tier. Training the head on cached features is cheap. The full ablation campaign of 27 federated runs, each of 20 rounds, completed on the same machine. The recurring cost of a real deployment would be the hospitals' own compute and a coordinator server; since each update is about 1 MB, bandwidth is not a significant cost. The measured overhead is reported in Section 7.9.

3.2.3 Social Feasibility

The system is aimed at a real obstacle to medical AI, the inability to share patient images, and it keeps patient control of data as a design invariant. It is presented as a decision-support and triage aid whose uncertain outputs are referred to a clinician, not as a replacement for one. Two risks are acknowledged. A model trained on public data may not transfer to a particular hospital's population, and a screening tool with limited sensitivity can give false reassurance. Both are reported with measured numbers in Chapter 7, and the application states that it is a research prototype.

3.3 System Specification

3.3.1 Hardware Specification

Table 3.1 Development hardware

Component	Specification
GPU	NVIDIA GeForce RTX 3050 Laptop GPU, 4 GB VRAM
CPU	16 logical cores
Memory	14 GB RAM
Storage	Local SSD; raw data, caches and checkpoints occupy about 64 GB
Operating system	Linux, with Docker for the multi-container demonstration

3.3.2 Software Specification

Table 3.2 Principal software and pinned versions

Layer	Software	Version
Language	Python	3.11
Deep learning	PyTorch / torchvision	2.13.0 / 0.28.0
Federated learning	Flower (flwr)	1.35.0
Differential privacy	Opacus	1.6.0
Preprocessing	opencv-python-headless	5.0.0.93
Explainability	grad-cam	1.5.7
Configuration	Hydra / OmegaConf	1.3.5 / 2.3.1
Tracking	MLflow (local, SQLite)	3.15.2
Metrics and OOD	scikit-learn	1.9.0
Data handling	NumPy / pandas / pydicom / Pillow	2.4.6 / 2.3.3 / 3.0.2 / 12.3.0
Demo application	Streamlit with a React component	1.62.0
Testing	pytest	9.1.1
Containers	Docker Compose	CPU-only images

CHAPTER 4

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The system is a star topology. Every hospital runs an identical local pipeline, and the server only aggregates and redistributes (Figure 4.1). A hospital loads its own X-rays, enhances contrast with CLAHE, resizes and normalises them, and passes them through the frozen DenseNet121 backbone. Only the small classifier head is trained, using DP-SGD. The head's parameters are quantised and masked by SecAgg+ and sent over a TLS channel on which the hospital has authenticated itself. The server removes the masks from the aggregate only, applies FedAvg, and broadcasts the new head. The invariant that the whole design protects is that the only thing leaving a hospital is a small, noised, masked update sent over an authenticated encrypted channel.

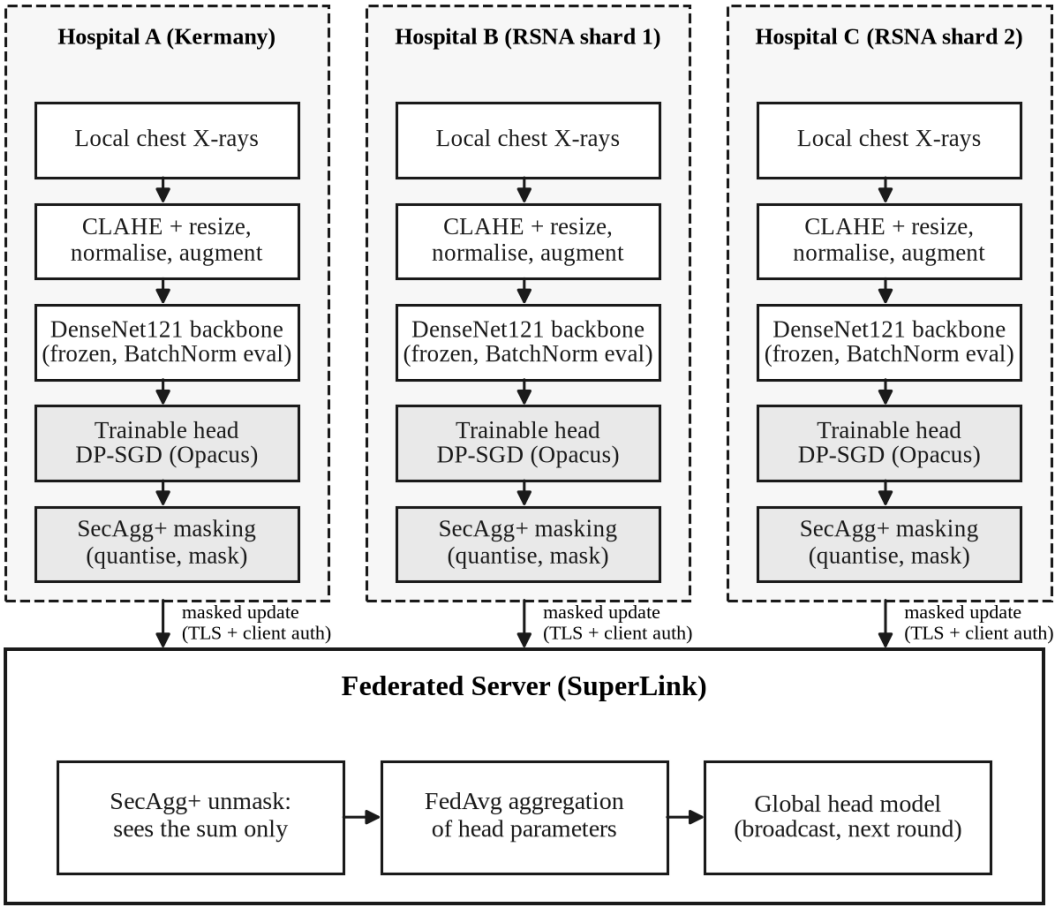


Figure 4.1 Architecture of the federated pipeline

The design rests on seven recorded decisions. The most important is to freeze the backbone with its BatchNorm layers in evaluation mode and to federate only the head. A frozen BatchNorm is a fixed affine transform, so it is safe for per-sample gradients, no running statistics are averaged across non-IID hospitals, and the update shrinks from

about 7 million to about 263 thousand parameters. The cost is a ceiling on attainable accuracy relative to full fine-tuning, which is measured in Section 7.8. The other decisions are summarised in Table 4.1.

Table 4.1 Principal design decisions

Decision	Choice	Reason
Trainable part	Frozen backbone, trainable head	DP compatibility, update size, SecAgg cost, 4 GB VRAM
Privacy unit	Sample-level DP-SGD with RDP accountant	The claim is about patients, not hospitals; ε and δ must be reported
Secure aggregation	Flower SecAgg+ only	No custom cryptography
Transport	TLS with per-hospital client authentication	Server-only TLS does not stop a fake hospital connecting
Flower version	Pinned exactly (1.35.0)	API changed between releases
Preprocessing	OpenCV for CLAHE only, explicit BGR to RGB	Mis-ordered channels silently cost accuracy
Data separation	Patient-level splits	Image-level splits leak patients across train and test
Execution	One client code path, two engines	Simulation for measurement, deployment for demonstration

Threat model. The adversaries considered are an honest-but-curious server, a passive network observer, an unregistered party impersonating a hospital, and collusion among clients up to the SecAgg+ threshold. Malicious clients that submit poisoned updates are out of scope. This exclusion is not an oversight: secure aggregation exists to stop the server inspecting individual updates, and detecting poisoned updates requires inspecting them, so the two aims are directly opposed and need dedicated techniques to reconcile. Table 4.2 maps each layer to the asset it protects.

Table 4.2 Protection layers and what each defends against

Layer	Protects	Against	Mechanism
Federated learning	Raw images	Any party outside the hospital	Images stay in local storage
Differential privacy	Information inside an update	Inference from updates	DP-SGD, (ε, δ)
Secure aggregation	A hospital's individual update	The server itself	SecAgg+ masking
TLS + client auth	Messages in transit	Eavesdropping, tampering, impersonation	gRPC, mutual authentication

A noted limitation follows from the choice to add noise independently at each hospital. This is effectively local differential privacy, which has lower utility than central differential privacy at the same ε. The principled extension is distributed differential privacy, where each hospital adds a share of the noise and secure aggregation reveals only the sum, but it requires discrete noise over a finite field integrated with the aggregation

protocol and is left as future work.

4.2 Design

4.2.1 Data Flow Diagram

Figure 4.2 shows the main data flows. Raw images flow only from the hospital to its own preprocessing and local training processes. What crosses to the aggregation process is the masked update, and what returns is the global head. The inference process, used by a clinician, receives the global model and an uploaded image, and returns the result, heatmap and confidence.

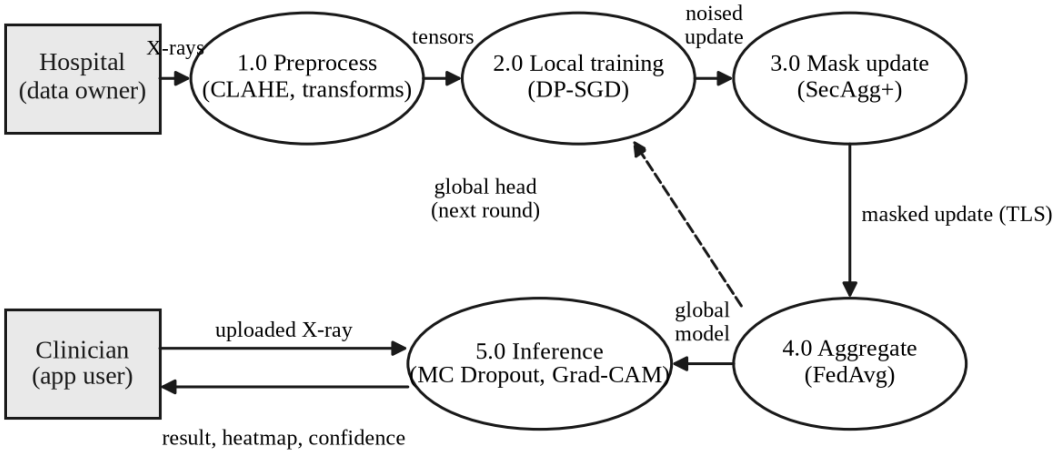


Figure 4.2 Level-1 data flow diagram

4.2.2 Use Case Diagram

Three actors interact with the system (Figure 4.3). The hospital administrator starts local training and the sending of masked updates. The federated server aggregates and broadcasts. The clinician uploads an X-ray, reads the prediction and confidence, inspects the explanation, and reviews cases that the system has deferred.

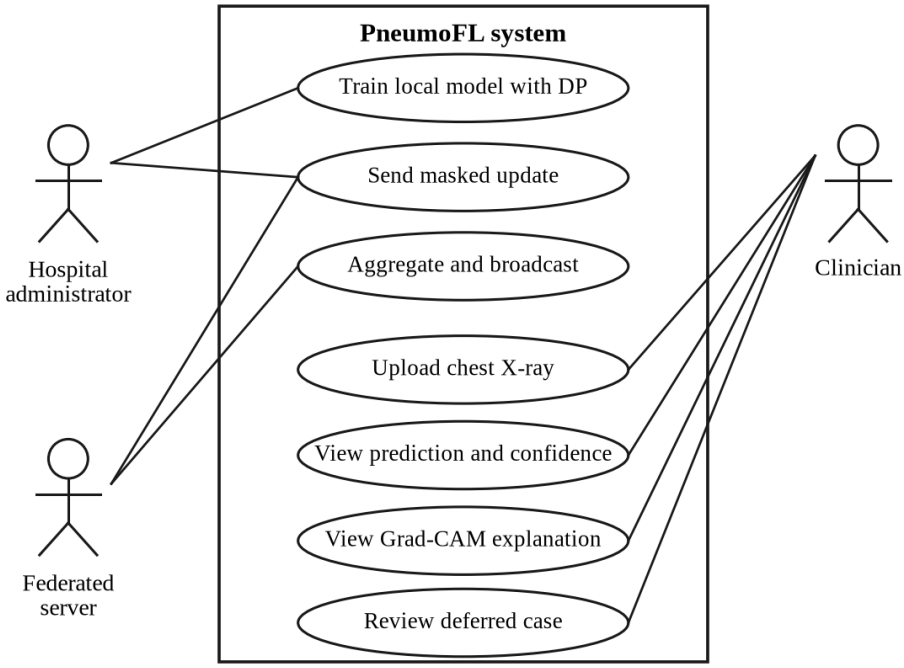


Figure 4.3 Use case diagram

4.2.3 Class Diagram

The main classes are shown in Figure 4.4. The head wraps the frozen backbone; the hospital client trains it through the DP trainer and communicates with the server; the Monte Carlo Dropout predictor and the Grad-CAM explainer operate on the global model at inference time.

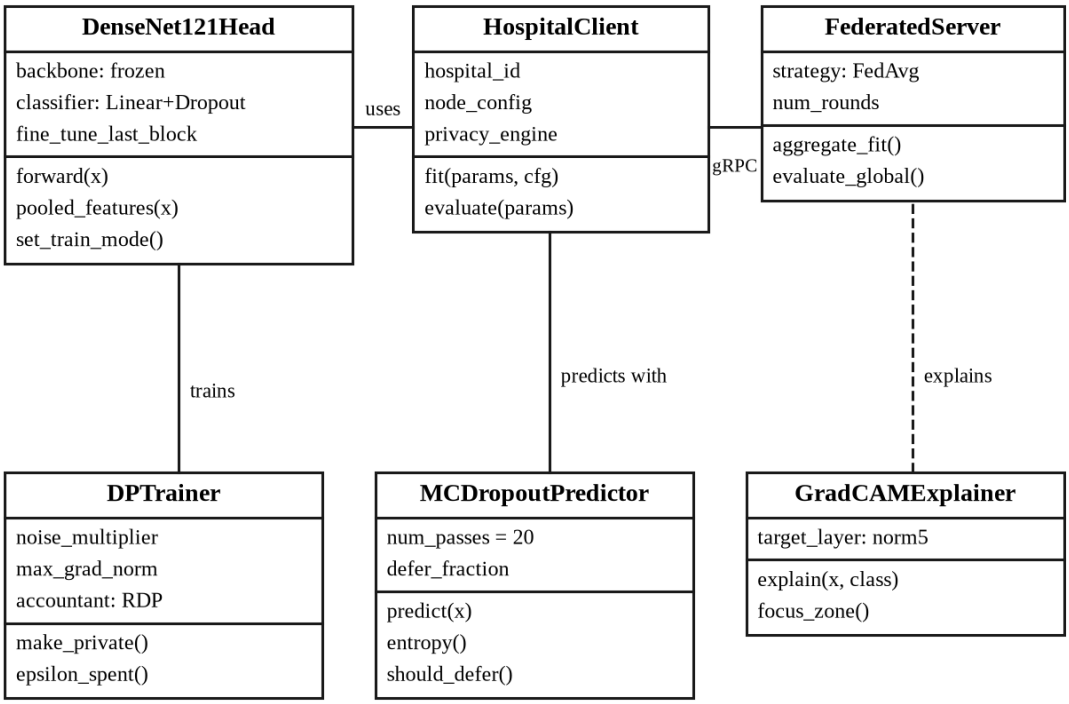


Figure 4.4 Class diagram of the principal modules

4.2.4 Sequence Diagram

One federated round is shown in Figure 4.5. The server broadcasts the global head, each hospital trains locally under DP-SGD, quantises and masks the result, and sends it over the authenticated channel. The server cancels the masks in the aggregate, applies FedAvg, and logs the round's metrics.



Figure 4.5 Sequence diagram of one federated round

CHAPTER 5

5. METHODOLOGY AND TESTING

This chapter describes the pipeline module by module, in the order a sample travels through it, and then the testing carried out on it. The module description records the parameters actually used.

5.1 Module Description

5.1.1 Data Acquisition and Partitioning

Two sources are used. The Kermany chest X-ray set (5,856 images from 3,081 patients, 73 percent pneumonia) is downloaded from its Mendeley Data release and verified against the published SHA-256 checksum. The RSNA Pneumonia Detection data (26,684 patients, 22.5 percent pneumonia) is obtained from Kaggle. Kermany's filenames encode an accession identifier, so the 5,856 images could be grouped by patient, and RSNA carries a patient identifier directly. All splits into train, validation and test are therefore made at the patient level, and tests verify that no patient appears in two splits.

Kermany forms Hospital A and the RSNA patients are divided into two equal shards of 13,342 patients that form Hospitals B and C. This is the natural partition, and the three sites genuinely differ in age group, scanner and prevalence. Because Hospital A is about four and a half times smaller than B and C, a size-balanced companion partition was also built by stratified subsampling of B and C down to the size of A. A third, synthetic regime uses a Dirichlet split with concentration α to control label skew. The partition indices are committed as JSON, so a third party can reproduce the exact splits.

RSNA marks each patient as Normal, Lung Opacity, or No Lung Opacity / Not Normal. Following the dataset's own two-class target, the last category (an abnormal film without pneumonia) is grouped with Normal. This keeps the full negative class but means the model is taught that "abnormal but not pneumonia" equals "normal". The consequence for the false-positive rate is measured in Section 7.4.

5.1.2 Preprocessing

Each image is decoded (pydicom for DICOM files), converted explicitly from the BGR order OpenCV returns to RGB, and enhanced with CLAHE using a clip limit of 2.0 and an 8 by 8 tile grid. The result is cached to disk so that preprocessing is neither a source of run-to-run variation nor a throughput bottleneck. torchvision then resizes to 224 by 224 pixels and normalises with ImageNet statistics. The training transform adds a random rotation of up to 10 degrees and a brightness and contrast jitter of 0.1. OpenCV is used for CLAHE only, because torchvision does not provide it and everything else composes better and seeds more cleanly in torchvision.

5.1.3 Model

The model is DenseNet121 with ImageNet weights. The backbone is frozen and its BatchNorm layers are held in evaluation mode, so their running statistics never change. The 1024-dimensional globally pooled feature vector feeds a head consisting of a linear layer from 1024 to 256 units, a ReLU, a dropout layer with rate 0.3, and a linear layer to two outputs, 262,914 trainable parameters in all. The dropout placement is a deliberate choice, because stock DenseNet121 has no dropout at all. Placing it in the head keeps the backbone and head separation clean at the price that Monte Carlo Dropout captures only last-layer uncertainty. Class imbalance is handled with an inverse-frequency weighted loss. Since the backbone is fixed, its pooled features are computed once and cached, and training the head then costs seconds.

5.1.4 Federated Training

Flower orchestrates the rounds. In each round every hospital receives the global head, trains it for one local epoch with Adam (learning rate 0.001, batch size 32), and returns the new parameters, and the server combines them with FedAvg weighted by sample count. A run uses 20 rounds, and the model reported is the one chosen by pooled validation AUROC, never by the test set. The same client and server code runs under Flower's simulation engine, where clients execute one after another on the single GPU, and under its deployment engine in separate containers.

5.1.5 Differential Privacy

When privacy is enabled, the local step uses Opacus DP-SGD. Each per-sample gradient is clipped to norm C = 1.0 and Gaussian noise is added to the clipped sum before the update (Equation 5.1). For a batch of size B the noisy gradient is

$$\tilde{g} = (1 / B) \cdot (\Sigma_i g_i \cdot \min(1, C / \|g_i\|) + N(0, \sigma^2 C^2 I)) \quad (5.1)$$

The noise multiplier σ is computed from the target ε (4 by default, swept over 1, 2, 4 and 8), $\delta = 10^{-5}$, the dataset size, the batch size and the total number of steps, using the RDP accountant. The value of δ is well below 1/N for every hospital (the smallest has 4,180 training images). Each hospital keeps one privacy engine for the whole run, so the accountant accumulates the spent budget across rounds. DP training uses the deterministic, non-augmented view of each image, since augmentation would change the sensitivity analysis. Adding noise without clipping and accounting is explicitly prohibited, because it carries no formal guarantee.

5.1.6 Secure Aggregation

Secure aggregation uses Flower's SecAgg+ workflow with 3 secret shares and a reconstruction threshold of 2. A hospital quantises its update to integers, adds pairwise masks and sends the masked vector, and the server learns only the sum. Quantisation interacts with accuracy and with the DP noise, so its effect is measured as its own row of the ablation. No cryptographic code was written for this project. At the pinned Flower version SecAgg+ is available only through the older strategy and workflow interface, so a separate client and server application pair was built on that interface, and the main application was left untouched.

5.1.7 Transport Security

The server runs with TLS using a project certificate authority, and each hospital's SuperNode holds its own ECDSA key pair whose public half is registered with the server beforehand. A node that is not registered is refused. Certificate and key generation is scripted with openssl and ssh-keygen, and the generated material is excluded from version control. A check confirms that the real payload fits comfortably within Flower's gRPC message limit.

5.1.8 Explanation, Uncertainty and Input Checks

Grad-CAM. The explanation uses the pytorch-grad-cam library on the last feature layer of the backbone. Because the whole backbone is frozen, nothing in the forward pass requires gradients and the library's backward hook never fires, so the input tensor is explicitly marked as requiring gradients. This was found by a crash during the first live test, not by inspection.

Monte Carlo Dropout. The cached pooled feature of an image is passed through the head T = 20 times with dropout active, and the mean of the softmax outputs gives the prediction. Uncertainty is the entropy of that mean (Equation 5.2):

$$H(\bar{p}) = -\sum_c \bar{p}_c \log \bar{p}_c, \quad \bar{p} = (1/T) \sum_t \mathrm{softmax}(z_t) \tag{5.2}$$

The deferral policy is a fixed coverage target. The 10 percent of predictions with the highest entropy are referred to a clinician, and the entropy cutoff is derived from the validation distribution of each model, not hand-picked.

Calibration. A temperature T is fitted on the validation set so that the confidence matches the observed accuracy (softmax(z / T)). The decision threshold τ on P(pneumonia) is chosen on the validation set only, from a sweep of 0.10 to 0.90, as the value with the best F1. A band around τ is reported as "inconclusive" instead of forcing a label.

Out-of-distribution check and input gate. One isolation forest per hospital is fitted to the pooled features of that hospital's images with a 5 percent flag target, to indicate that an image is unlike the training data. Separately, a small supervised gate rejects images that are not chest X-rays at all. Two earlier gate designs (colour saturation, and a classifier trained on synthetic non-X-ray images) were tried and discarded because

they failed on real photographs.

5.1.9 Evaluation, Tracking and the Demonstration Application

The metrics module computes AUROC (primary), AUPRC, sensitivity, specificity, precision, F1, balanced accuracy and the confusion matrix, with bootstrap 95 percent confidence intervals on AUROC. Every federated run is logged to a local MLflow store, including the resolved configuration, and the results tables are generated by querying that store, not copied by hand. An overhead module records bytes sent per client per round and wall-clock time per round. The demonstration application is a Streamlit page with a React front end. It accepts an image, applies the input gate and quality checks, runs the classifier with Monte Carlo Dropout, and displays the label, confidence, the Grad-CAM overlay (also on a 3D lung model) and the deferral state. It does not change training or any privacy layer.

5.2 Testing

Automated tests were written specifically where a silent failure would be invisible. The suite contains 255 tests, and all of them pass. Table 5.1 maps each requirement to the checks that cover it.

Table 5.1 Test coverage of the main risks

Area	What is verified
Partitioning	No patient overlap between train, validation and test; partitions are deterministic for a given seed
Preprocessing	BGR to RGB conversion is correct; CLAHE is deterministic for fixed parameters
Differential privacy	Clipping and noise match the intended mechanism; the accountant consumes budget across rounds
Secure aggregation	Masks cancel exactly: the unmasked aggregate equals plain FedAvg within tolerance
Transport	Unregistered node is rejected; registered node is accepted; payload fits the message limit
Model	Backbone stays frozen and BatchNorm stays in eval mode; head passes Opacus validation
Explanation, uncertainty	Grad-CAM runs with a frozen backbone; dropout is actually active in repeated passes; deferral hits its coverage
Metrics and calibration	Metric values, bootstrap intervals, calibration error and conformal coverage on known inputs
Application	Inference wrapper, presentation payload, front-end bundle presence
Integration	A 2-client, single-round federated run completes end to end

Testing against real runs, and not only synthetic inputs, uncovered several defects that unit tests alone would have missed. A multi-source hospital loader silently dropped data when one hospital held images from two datasets; a Dirichlet client-naming mismatch broke one sweep; the first launch of the ablation campaign was corrupted by a

missing flag; and the integration smoke test was found to be contaminating the MLflow store with its own runs. Opacus's per-sample hooks crashed on in-place ReLU layers inside the last dense block the first time that block was trained under DP. In the container deployment, a CUDA dependency chain bloated the build, a TLS certificate lacked the server's container hostname, and a registration race left a hospital unregistered when it first connected. All were fixed and none changed a reported result. A further apparent regression, in which secure aggregation seemed to stop working, turned out to be the harness omitting the number of simulated nodes, because the framework defaults to two and waited for a third that never started.

CHAPTER 6

6. PROJECT DEMONSTRATION

The system was demonstrated in three ways: in simulation for measurement, as a multi-container deployment with real security, and through the web application.

6.1 Simulation Runs

A federated run is started with Flower's simulation engine and three simulated SuperNodes, for example with differential privacy at ε = 4. The run prints the pooled validation and test AUROC each round and the privacy budget spent so far, and logs them to MLflow. The ablation campaign was executed this way: 27 federated runs (three seeds each of FedAvg on two partitions, FedAvg with SecAgg+, four DP budgets, and two Dirichlet settings), all of which completed.

6.2 Multi-Container Deployment

The deployment uses Docker Compose with one SuperLink container and three hospital containers, CPU-only. Certificates and node keys are generated by script, each hospital container mounts only its own shard of the data, and all three hospitals participated in every round. A three-round run completed with real TLS and client authentication. A negative test confirmed that a SuperNode whose key had not been registered was refused, and a positive test confirmed that a registered one was accepted.

The combination of all layers (FedAvg with SecAgg+, DP and TLS with client authentication) was run once in simulation and once in the containers. In simulation the pooled test AUROC rose from 0.754 through 0.784 to 0.801 over three rounds with a spent budget of ε = 3.37 against a target of 4. In the container run, two rounds completed in 151 seconds with ε near 3.6. This was a demonstration that the combination works, not a multi-seed measurement.

6.3 Web Application

The model is also served through a web application at pneumonia-fl-demo.streamlit.app. A user uploads a JPEG, PNG or DICOM chest X-ray, the application checks image quality and rejects anything that is not a chest X-ray, and the analysis runs the Monte Carlo Dropout classifier, the out-of-distribution check and Grad-CAM. The result page shows the finding, a confidence value, a certainty band, whether the case is flagged for a second look, and the heatmap, both on the film and on a 3D lung model. The model behind the application is the differentially private federated model (ε = 4), so a visitor sees the privacy-preserving system itself. An example of the explanation images the module produces is shown in Figure 6.1.

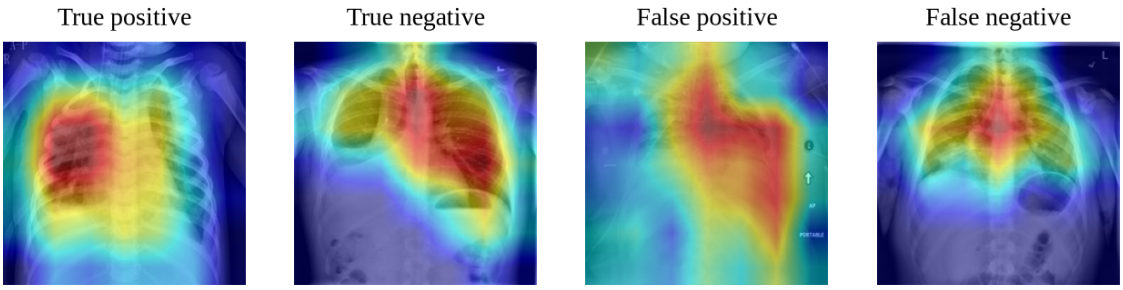


Figure 6.1 Grad-CAM overlays for a true positive, true negative, false positive and false negative

The heatmaps should be read with care. Section 7.7 shows that localisation against expert boxes is modest, so the overlay is an aid for the clinician to check, not evidence that the model attended to the right lesion.

6.4 Decision Policy of the Deployed Model

The deployed model does not use the default threshold of 0.5. At 0.5 the differentially private model is biased toward the normal class, with a validation sensitivity of only 44.7 percent. The threshold is therefore taken from a validation sweep. The model's probabilities are first scaled with the fitted temperature ($T = 2.2534$), and a decision threshold of 0.35 is then chosen as the best-F1 point on those calibrated validation probabilities. A band of plus or minus 0.05 around the threshold is reported as inconclusive, and the 10 percent of cases with the highest entropy (cutoff 0.673) are flagged for review.

An error found late in the work is worth recording. An earlier threshold of 0.2 had been selected on the raw probabilities but was applied by the application to the temperature-scaled ones, a mismatch that pushed the live behaviour to about 89 percent sensitivity and 47 percent specificity on the test set. Re-deriving the threshold on the same scale on which it is applied gave 0.35, and the figures in Chapter 7 are for the corrected policy.

CHAPTER 7

7. RESULT AND DISCUSSION (COST ANALYSIS)

7.1 Evaluation Protocol

All numbers are measured on the held-out pooled test set (4,838 images, 31.0 percent pneumonia) of the natural partition unless stated otherwise. Results are the mean and standard deviation over three seeds (42, 123 and 2024). Each federated run uses 20 rounds, and the privacy parameter is reported with $\delta = 10^{-5}$. Model selection, the decision threshold, the temperature and the deferral cutoff use the validation split only. Two operating points appear in this chapter and are labelled: a fixed threshold of 0.5, which is how the ablation ladder compares configurations, and the deployed threshold of 0.35 with calibration.

7.2 The Ablation Ladder

Table 7.1 gives the cost of each layer in AUROC. Centralized training on the pooled data, with no privacy, is the ceiling at 0.905. Moving to federated averaging lowers AUROC to 0.814, a drop of about nine points that reflects both the frozen-backbone capacity limit and genuine heterogeneity between the hospitals. Secure aggregation is almost free in accuracy: 0.819, which is indistinguishable from FedAvg given the standard deviations. Differential privacy at $\varepsilon = 4$ gives 0.809.

Table 7.1 Ablation ladder, pooled test AUROC (mean ± std over 3 seeds)

#	Configuration	Regime	AUROC
1	Local, per-hospital average (floor)	natural	0.8924 ± 0.0005
2	Centralized pooled (ceiling)	natural	0.9053 ± 0.0006
2	Centralized pooled (ceiling)	balanced	0.9290 ± 0.0010
3	FedAvg	natural	0.8144 ± 0.0095
3	FedAvg	balanced	0.8387 ± 0.0273
4	FedAvg + Secure Aggregation	natural	0.8194 ± 0.0200
5	FedAvg + DP, ε = 1	natural	0.7909 ± 0.0147
5	FedAvg + DP, ε = 2	natural	0.8021 ± 0.0122
5	FedAvg + DP, ε = 4	natural	0.8085 ± 0.0097
5	FedAvg + DP, ε = 8	natural	0.8133 ± 0.0080
6	Full system (FedAvg + SecAgg+ + DP + TLS)	natural	demonstrated once, not a 3-seed row

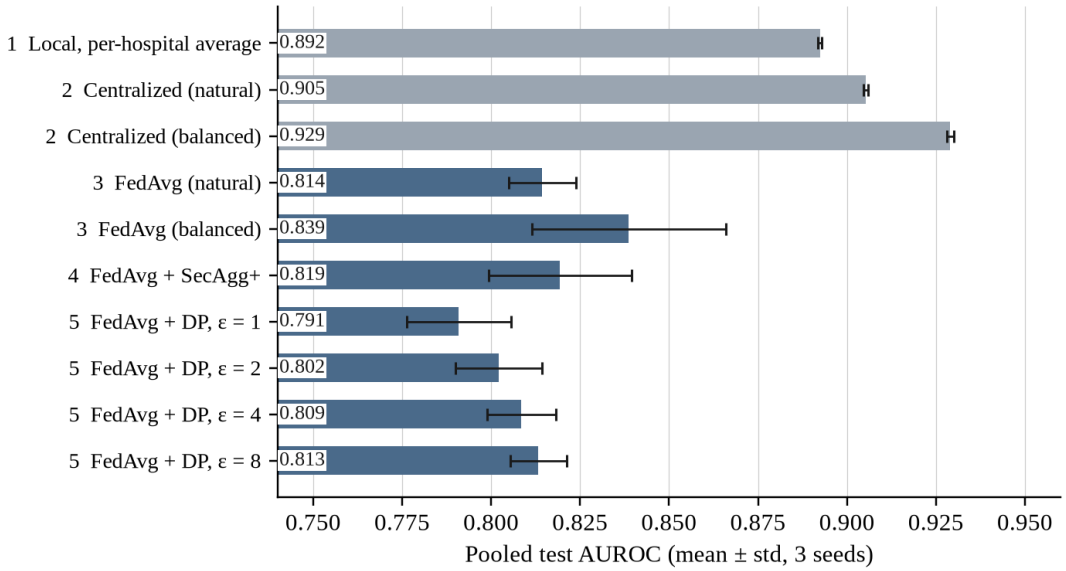


Figure 7.1 Ablation ladder across configurations

The expected result that federation would beat the best single hospital did not hold in this data. The local models average 0.892, above the federated 0.814. The most likely reason is the label harmonisation described in Section 5.1.1: Hospitals B and C see a harder boundary, with abnormal non-pneumonia films labelled normal, than Hospital A, so averaging partly mixes incompatible boundaries instead of reconciling them. This is reported as found. The measured value of federation here is that it is possible without moving images, and its cost is quantified, not that it outperforms local training.

7.3 Privacy-Utility Trade-off

Across the sweep from ε = 1 to ε = 8, AUROC rises monotonically from 0.791 to 0.813 (Figure 7.2), with no reversals. The curve is shallow, which is the main benefit of training only a head: noise is spread over about 263 thousand parameters instead of 7 million. The step from ε = 4 to ε = 8 gains only 0.005 AUROC, less than the seed-to-seed standard deviation of about 0.01, so the default budget of 4 gives up nothing that can be measured relative to 8 and gives a stronger guarantee.

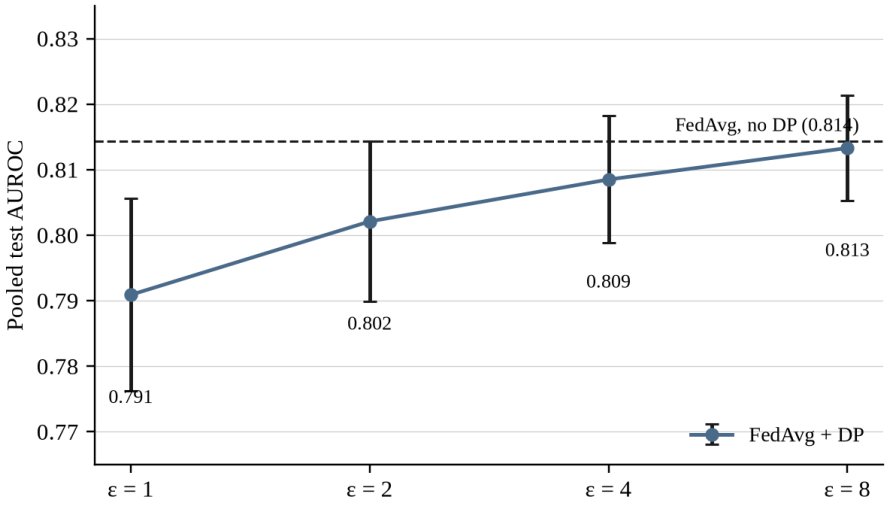


Figure 7.2 Privacy-utility curve over the epsilon sweep

AUROC hides which errors DP introduces. Table 7.2 shows the full classification breakdown at the default threshold of 0.5.

Table 7.2 DP sweep at threshold 0.5 (mean ± std over 3 seeds)

ε	AUROC	Accuracy	Sensitivity	Specificity	F1
1	0.7909 ± 0.0180	0.7463 ± 0.0124	0.4569 ± 0.0243	0.8766 ± 0.0078	0.5279 ± 0.0254
2	0.8021 ± 0.0149	0.7523 ± 0.0119	0.4698 ± 0.0198	0.8795 ± 0.0084	0.5408 ± 0.0224
4	0.8085 ± 0.0119	0.7542 ± 0.0105	0.4763 ± 0.0178	0.8793 ± 0.0073	0.5460 ± 0.0198
8	0.8133 ± 0.0097	0.7589 ± 0.0096	0.4896 ± 0.0154	0.8801 ± 0.0070	0.5576 ± 0.0176

Specificity stays near 0.88 throughout, while sensitivity sits between 0.46 and 0.49. Noise therefore costs sensitivity, not specificity, and at the default threshold the model misses more than half of the pneumonia cases. This is what makes the threshold selection of Section 6.4 necessary and shows that an AUROC figure alone understates the clinical cost of privacy.

7.4 Performance of the Deployed Model

Table 7.3 reports the deployed operating point (temperature 2.2534, threshold 0.35) applied to the three seeds' checkpoints. The temperature was fitted on the seed-42 validation set and kept fixed for the other seeds, so the standard deviations include that mismatch. The sensitivity and specificity are balanced at about 73 percent.

Table 7.3 Deployed policy over 3 seeds, pooled test set (mean ± std)

Metric	Value
AUROC	0.7996 ± 0.0121
AUPRC	0.6176 ± 0.0247
Sensitivity	0.7392 ± 0.0109
Specificity	0.7317 ± 0.0185
Precision	0.5540 ± 0.0163
NPV	0.8617 ± 0.0048
F1	0.6332 ± 0.0106
Accuracy	0.7340 ± 0.0118
Balanced accuracy	0.7355 ± 0.0087
ECE after calibration	0.0258 ± 0.0007
Brier score after calibration	0.1639 ± 0.0048

The AUROC here, 0.800, is lower than the 0.809 of Table 7.1 for the same checkpoints. The ablation scores a single deterministic pass, while the application averages 20 dropout passes, and the two are separate measurements. For the seed-42 checkpoint that is actually deployed, Table 7.4 gives the full set of metrics with a confusion matrix, and also a column with the ambiguous RSNA group excluded.

Table 7.4 Deployed checkpoint (seed 42, threshold 0.35)

Metric	All test cases (n = 4,838)	Ambiguous group excluded (n = 3,077)
AUROC (95% CI)	0.787 (0.774–0.800)	0.884 (0.872–0.896)
AUPRC	0.599	0.892
Sensitivity	73.1%	73.1%
Specificity	72.2%	89.0%
Precision	54.2%	86.3%
NPV	85.6%	77.6%
F1	0.623	0.792
Accuracy	72.5%	81.2%
Balanced accuracy	72.7%	81.0%
TP / FP / FN / TN	1098 / 927 / 404 / 2409	1098 / 174 / 404 / 1401

The ambiguous group. The second column removes 1,761 RSNA images that the dataset labels Normal but which show some other abnormality, such as scarring or fluid. The model flags 42.8 percent of these as pneumonia, against 9.3 percent of the RSNA images that are truly normal, so 753 of the 927 false positives come from this one group. Removing it leaves sensitivity untouched, because no pneumonia case is removed, and raises specificity and precision. This does not make the model better. It is the same model scored on an easier subset. The full-set column is the figure that describes use on patients who may have any finding, and the filtered column explains why that figure is lower than

the model deserves. A supplementary experiment then retrained the head-only FedAvg + DP model (ε = 4, 20 rounds, natural regime, seeds 42, 123 and 2024) with the ambiguous group removed from every training split, and scored both models on the same original test set at threshold 0.5 without calibration. On the clear-label cases AUROC rose from 0.907 ± 0.007 to 0.930 ± 0.006, so retraining was not fully redundant with excluding the group at evaluation time. On the full test set it fell from 0.809 ± 0.012 to 0.780 ± 0.020, and specificity dropped from 0.88 to 0.62, because a model that never saw such images calls them pneumonia. The filtered model is therefore not adopted, the label harmonization and the deployed model are unchanged, and the result is not comparable with the ablation table. It bounds the limitation: most of the gap between the full-set and clear-label figures is label ambiguity that retraining does not recover.

7.5 Calibration, Deferral and Conformal Coverage

Without differential privacy the Monte Carlo Dropout confidence is reasonably calibrated (ECE about 0.02 to 0.03). Switching DP on roughly quadruples the calibration error, to between 0.094 and 0.104 across the sweep, and the error is flat over ε, so it is caused by the clipping and noise process itself, not by the amount of noise. Table 7.5 and Figure 7.3 give the values. Temperature scaling repairs most of this: for the deployed policy ECE falls from about 0.10 to about 0.03 over the three seeds.

Table 7.5 Calibration error by configuration (3 seeds)

Configuration	ECE	Brier score
Centralized	0.0262 ± 0.0053	0.1182 ± 0.0007
FedAvg	0.0227 ± 0.0070	0.1800 ± 0.0066
FedAvg + SecAgg	0.0287 ± 0.0169	0.1875 ± 0.0141
FedAvg + DP, ε = 1	0.0943 ± 0.0079	0.1805 ± 0.0065
FedAvg + DP, ε = 2	0.0998 ± 0.0040	0.1762 ± 0.0058
FedAvg + DP, ε = 4	0.1023 ± 0.0037	0.1744 ± 0.0043
FedAvg + DP, ε = 8	0.1038 ± 0.0059	0.1729 ± 0.0041

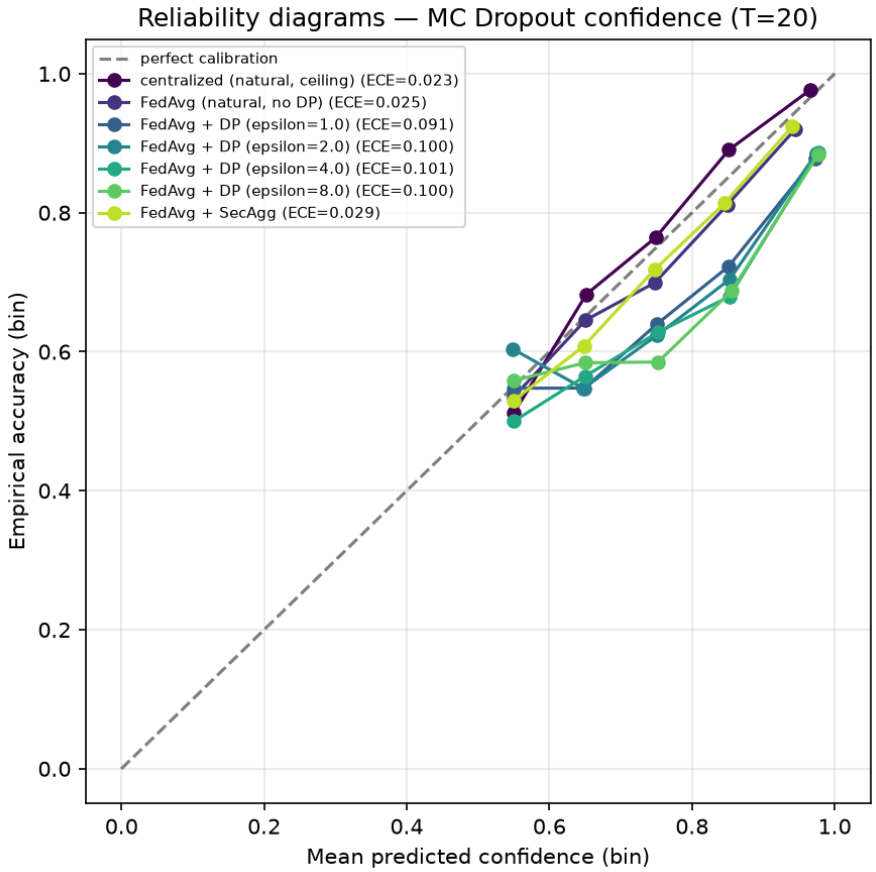


Figure 7.3 Reliability diagrams by configuration

The more clinically important property survives intact. The risk-coverage curves rise monotonically with coverage in every configuration, so the cases the model is least certain about really are the ones most likely to be wrong, and the deferral ranking works even where the numeric confidence is distorted. On the pooled test set, the 10 percent of cases deferred by the deployed policy are classified correctly 48.7 percent of the time (chance level), against 75.2 percent for the remaining cases. Table 7.6 shows the effect of the two review layers on the seed-42 deployed model.

Table 7.6 Effect of abstention and deferral, deployed checkpoint

Stage	Cases kept	Accuracy	Sensitivity	Specificity
All cases	100%	72.5%	73.1%	72.2%
After inconclusive band (12.7% abstained)	87.3%	75.2%	76.1%	74.8%
After abstention and deferral (10.1% deferred)	77.1%	78.7%	70.7%	81.8%

Split conformal prediction, calibrated on the validation set for 90 percent coverage, achieves an empirical test coverage of 0.908 for the $\varepsilon = 4$ model (mean set size 1.36), so the coverage guarantee holds even for the configurations whose raw confidence is distorted. The isolation-forest check flags 4.6 to 6.5 percent of in-distribution test images against its 5 percent target.

7.6 Empirical Privacy

A loss-based membership-inference attack was run against the checkpoints, with and without protection (Table 7.7). Every configuration sits within one percentage point of the chance value of 0.5, and the differentially private models, at 0.495 to 0.497, show no leakage.

Table 7.7 Membership-inference attack AUROC (3 seeds)

Configuration	Attack AUROC
Centralized, no protection	0.5084 ± 0.0011
FedAvg, no DP	0.4994 ± 0.0001
FedAvg + DP, ε = 1	0.4952 ± 0.0021
FedAvg + DP, ε = 2	0.4959 ± 0.0019
FedAvg + DP, ε = 4	0.4966 ± 0.0018
FedAvg + DP, ε = 8	0.4970 ± 0.0018

This result must be read carefully, and it is not evidence that differential privacy is unnecessary. The attack finds almost nothing even in the unprotected centralized model, which is consistent with low memorisation by a frozen backbone and a small head, but is equally consistent with the attack being too weak to find leakage that a stronger method, such as a shadow-model attack or gradient inversion, might expose. The formal (ε, δ) guarantee is the privacy claim, and the attack result is a sanity check that no obvious leak exists. A stronger attack is the natural next step.

7.7 Explanation Quality

Grad-CAM was scored against the RSNA expert bounding boxes with the pointing game (does the peak of the heatmap fall inside a box) and mean intersection over union. Localisation is modest in every configuration: the pointing-game accuracy is 0.184 for the centralized model, 0.163 for FedAvg and 0.147 for the ε = 4 model, with IoU of 0.099, 0.095 and 0.084. Federation and privacy lower it slightly, but the differences lie within the variance across seeds. The weakness has a structural cause. With a completely frozen backbone, the classifier sees only a globally pooled 1024-number summary, so the heatmap is a reconstruction from correlated channels, not something the classifier used spatially. Figure 7.4 summarises the scores.

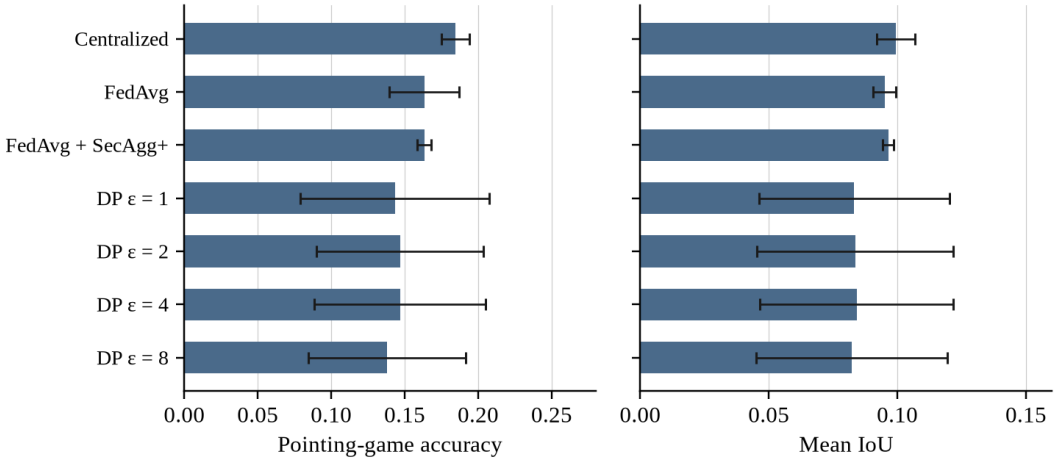


Figure 7.4 Grad-CAM localisation against expert boxes

7.8 Heterogeneity and the Fine-Tuning Fallback

The Dirichlet sweep isolates the effect of heterogeneity. With strongly skewed clients ($\alpha = 0.1$) FedAvg reaches 0.776, against 0.895 for nearly uniform clients ($\alpha = 1.0$). This swing is much larger than the cost of either secure aggregation or differential privacy, and it reproduces the known weakness of FedAvg under non-IID data.

The approved fallback of the freezing decision, unfreezing the last dense block with GroupNorm in place of BatchNorm, was evaluated over three seeds without privacy. Table 7.8 shows that it raises both the ceiling and the federated result. Combined with DP at $\varepsilon = 4$ in a single-seed first look, centralized training reached an AUROC of 0.863 with a sensitivity of 0.553, so the larger trainable part survives the noise, but sensitivity is again where privacy takes its toll. A federated run of the fine-tuned and private combination gave an AUROC of 0.830 and a sensitivity of 0.270 within a fixed round budget, with sensitivity still climbing. Neither combination has a multi-seed evaluation, so they are reported as indicative and are not part of the headline table.

Table 7.8 Frozen backbone versus fine-tuned last block, no privacy (3 seeds)

Configuration	Frozen AUROC	Fine-tuned AUROC
Centralized (ceiling)	0.9053	0.9251 ± 0.0004
FedAvg	0.8144	0.8651 ± 0.0101

7.9 Cost Analysis

The cost of the privacy layers shows up mainly in time and in accuracy, and hardly at all in bandwidth. Because only the head is exchanged, each client sends about 1.05 MB per round whether or not DP is on, against roughly 28 MB for a full DenseNet121, and this fits well within the gRPC limit. DP-SGD increases the wall-clock time of the final round from 0.52 s to 6.74 s (seed 42), an order of magnitude, because of Opacus's per-sample gradient computation. Secure aggregation costs almost no accuracy; its communication and compute overhead is expected to be larger but was not instrumented, because the overhead recording was wired into the main application only, which is a gap in the measurements. In accuracy terms, the cost is AUROC (0.905 to 0.814 for federation, then to 0.809 for DP at $\varepsilon = 4$), sensitivity at a fixed threshold, and a fourfold increase in calibration error that temperature scaling mostly removes. Table 7.9 collects these costs. In monetary terms the project used no paid services.

Table 7.9 Summary of the cost of each layer

Layer	Accuracy cost (AUROC)	Other costs
Federation (vs. centralized)	0.905 → 0.814	Heterogeneity sensitivity (0.776 at α = 0.1)
Secure aggregation	0.814 → 0.819 (no measurable loss)	Quantisation; masking handshake; overhead not instrumented
Differential privacy, $\varepsilon = 4$	0.814 → 0.809	Sensitivity at 0.5 falls to 0.48; ECE ×4; about 13× round time
TLS + client authentication	None	Certificate and key management scripts
Payload (all layers)	—	≈ 1.05 MB per client per round

7.10 Discussion and Limitations

The principal finding is the shape of the cost. Secure aggregation and TLS are close to free in accuracy, federation costs about nine AUROC points under this label scheme and data, and differential privacy adds a smaller further loss that hides a large loss of sensitivity. The architectural choice to train only a head is what keeps the privacy-utility curve shallow and the updates small, and its price is a ceiling on accuracy, which the fine-tuning study puts at roughly five AUROC points for the federated model.

The limitations are stated plainly. The hospitals are simulated on one machine. The differentially private training is effectively local DP, with worse utility than a distributed scheme. The ambiguous RSNA labels cap precision. Localisation of the explanation is weak, and confidence in the private model needs recalibration. The membership-inference test is too weak to be strong evidence. The deployed numbers rest on one seed's checkpoint, and the pooled test set comes from the same two datasets that the hospitals trained on, so there is no genuinely external validation. The full-system row has no multi-seed measurement. Malicious participants are not considered. Finally, the system is a research prototype: an AUROC near 0.8 with sensitivity and specificity near 73 percent is not a level at which it could be used for clinical decisions.

CHAPTER 8

8. CONCLUSION

This project set out to show that a pneumonia detector can be trained across hospitals that never share an image, with a formal privacy guarantee, a server that cannot see any single hospital's update, an authenticated encrypted channel, and predictions that come with an explanation and a confidence, and to measure what each of those layers costs. All of the layers were built from established components, with no custom cryptography, and they run together both in simulation and in a multi-container deployment with real TLS and client authentication. The work is backed by 255 automated tests and by a campaign of 27 seeded and tracked federated runs.

The measurements give a clear picture. Secure aggregation and authenticated transport cost almost nothing in accuracy. Federation costs about nine AUROC points against centralized training on this data. Differential privacy at $\varepsilon = 4$ adds a further small AUROC loss, but removes much of the sensitivity at the default threshold and quadruples the calibration error, so a threshold chosen on validation data and temperature scaling are needed to make the model usable. With those applied, the deployed model reaches an AUROC of about 0.80 with sensitivity and specificity near 73 percent on the full test set, and an AUROC of 0.88 when the images with ambiguous labels are set aside. Freezing the backbone is what makes the whole combination affordable, with a payload of about 1 MB per round and a shallow privacy-utility curve, and the fine-tuning study shows that its price is roughly five AUROC points in the federated setting.

Several results went against what had been hoped for, and they are reported as found. Federation did not beat the best single hospital, Grad-CAM localisation is weak, and the membership-inference attack was too weak to demonstrate the benefit of the privacy layer empirically. A mismatch between the scale on which the decision threshold was chosen and the scale on which it was applied was found and corrected late, which shows how easily a calibrated pipeline can be wrong without any test failing.

8.1 Future Work

The most useful next steps, in order, are:

- multi-seed measurements of the full combined system and of fine-tuning together with differential privacy, so that the sixth row of the ablation meets the same standard as the others and the recoverable accuracy is settled;
- distributed differential privacy, where hospitals add shares of the noise and the aggregation reveals only the total, which would remove the penalty of local noise;
- stronger privacy attacks (shadow-model membership inference and gradient inversion), an instrumented overhead measurement for secure aggregation, and an external validation set from a source unseen in training;
- defences against malicious participants, which need techniques that reconcile robust aggregation with secure aggregation.

CHAPTER 9

9. REFERENCES

Harvard format

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APPENDIX A – Sample Code

A.1 Preprocessing: CLAHE with explicit colour handling (src/data/preprocessing.py)

```
def apply_clahe(bgr_image: np.ndarray, params: ClaheParams) -> np.ndarray:
    """Apply CLAHE to the grayscale intensity, replicated back to 3 (identical)
        channels.

    Chest X-rays are grayscale; CLAHE is applied directly to intensity rather than via
    a color-space luminance channel, matching standard chest X-ray CLAHE usage in the
    literature — this project never applies CLAHE to a genuinely-color image.
    """
    gray = cv2.cvtColor(bgr_image, cv2.COLOR_BGR2GRAY)
    clahe = cv2.createCLAHE(clipLimit=params.clip_limit,
        tileGridSize=params.tile_grid_size)
    enhanced_gray = clahe.apply(gray)
    return cv2.cvtColor(enhanced_gray, cv2.COLOR_GRAY2BGR)
```

A.2 Preprocessing: BGR to RGB and the full per-image pipeline (src/data/preprocessing.py)

```
def preprocess_to_rgb(bgr_image: np.ndarray, params: ClaheParams) -> np.ndarray:
    """Full CLAHE pipeline: apply CLAHE, then the explicit BGR->RGB conversion."""
    enhanced_bgr = apply_clahe(bgr_image, params)
    return bgr_to_rgb(enhanced_bgr)
```

A.3 Noise multiplier from a target epsilon (src/privacy/accounting.py)

```
def compute_noise_multiplier(
    target_epsilon: float,
    target_delta: float,
    sample_rate: float,
    total_steps: int,
) -> float:
    """The Gaussian noise multiplier (sigma) that achieves target_epsilon after
    `total_steps` DP-SGD steps at the given `sample_rate` (batch_size / dataset_size),
    per the pinned Opacus version's RDP accountant."""
    return get_noise_multiplier(
        target_epsilon=target_epsilon,
        target_delta=target_delta,
        sample_rate=sample_rate,
        steps=total_steps,
        accountant="rdp",
    )
```

A.4 Monte Carlo Dropout predictive entropy (src/uncertainty/mc_dropout.py)

```
def predictive_entropy(mean_probs: torch.Tensor, eps: float = 1e-12) -> torch.Tensor:
    """Entropy of the MEAN predictive distribution across T passes (not the
    mean of T per-pass entropies) — the standard MC Dropout formulation."""
    return -(mean_probs * torch.log(mean_probs.clamp_min(eps))).sum(dim=-1)
```

A.5 Deferral by fixed coverage (src/uncertainty/deferral.py)

```
def compute_deferral(entropy: torch.Tensor, target_defer_fraction: float) ->
        DeferralResult:
    """Defers the `target_defer_fraction` highest-entropy predictions in this
    batch. `target_defer_fraction=0.10` defers the worst 10% by uncertainty."""
    if not 0.0 <= target_defer_fraction < 1.0:
        raise ValueError(f"target_defer_fraction must be in [0, 1), got
        {target_defer_fraction}")

    n = entropy.shape[0]
    num_defer = int(round(n * target_defer_fraction))
    if num_defer == 0:
        threshold = float(entropy.max().item()) + 1.0  # unreachable — nothing deferred
        deferred_mask = torch.zeros(n, dtype=torch.bool)
    else:
        sorted_entropy, _ = torch.sort(entropy, descending=True)
        threshold = float(sorted_entropy[num_defer - 1].item())
        # `>=`, not top-k indices directly: ties at the threshold value can
        # legitimately defer slightly more than `num_defer` examples — that is
        # correct (never silently under-defer a tied high-uncertainty case),
        # not a bug in the realized coverage.
        deferred_mask = entropy >= threshold

    coverage = 1.0 - deferred_mask.float().mean().item()
    return DeferralResult(threshold=threshold, deferred_mask=deferred_mask,
        coverage=coverage)
```